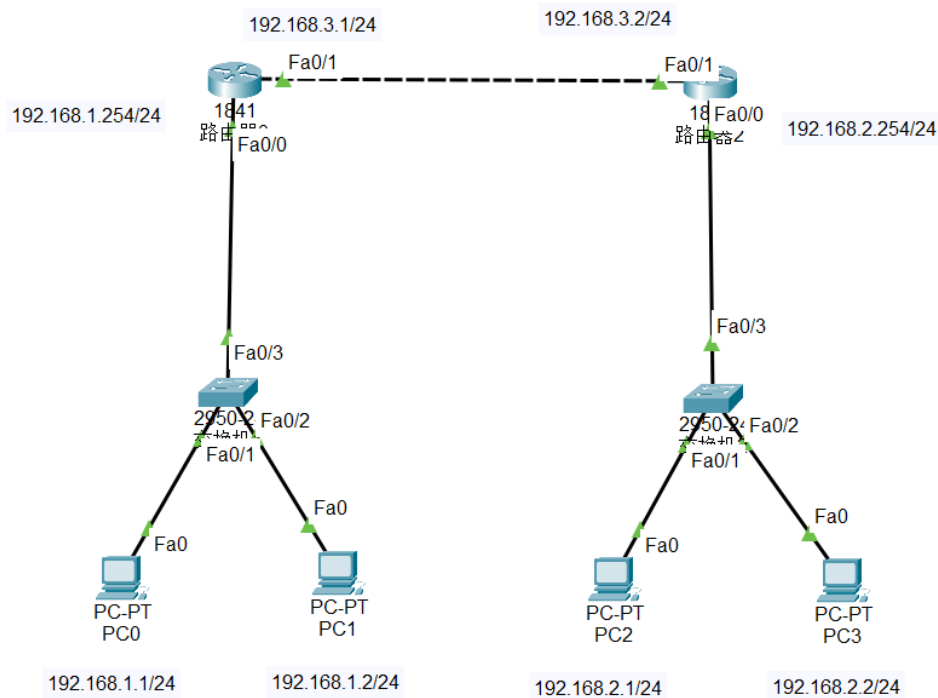


实验4



静态路由

R0

```
1 enable
2 configure terminal
3 ! 配置接口Fa0/0的IP地址（连接192.168.1.0/24网段）
4 interface FastEthernet0/0
5 ip address 192.168.1.254 255.255.255.0
6 no shutdown
7 ! 配置接口Fa0/1的IP地址（连接192.168.3.0/24网段）
8 interface FastEthernet0/1
9 ip address 192.168.3.1 255.255.255.0
10 no shutdown
11 ! 配置到192.168.2.0/24网段的静态路由，下一跳指向路由器1的Fa0/1接口（192.168.3.2）
12 ip route 192.168.2.0 255.255.255.0 192.168.3.2
13 end
14 write memory
```

R1

```
1 enable
2 configure terminal
3 ! 配置接口Fa0/0的IP地址（连接192.168.2.0/24网段）
4 interface FastEthernet0/0
5 ip address 192.168.2.254 255.255.255.0
6 no shutdown
7 ! 配置接口Fa0/1的IP地址（连接192.168.3.0/24网段）
8 interface FastEthernet0/1
9 ip address 192.168.3.2 255.255.255.0
```

```
10 no shutdown
11 ! 配置到192.168.1.0/24网段的静态路由，下一跳指向路由器0的Fa0/1接口（192.168.3.1）
12 ip route 192.168.1.0 255.255.255.0 192.168.3.1
13 end
14 write memory
```

RIP

R0

```
1 enable
2 configure terminal
3 ! 配置接口Fa0/0的IP地址
4 interface FastEthernet0/0
5 ip address 192.168.1.254 255.255.255.0
6 no shutdown
7 ! 配置接口Fa0/1的IP地址
8 interface FastEthernet0/1
9 ip address 192.168.3.1 255.255.255.0
10 no shutdown
11 ! 启用RIP协议，版本为RIP v2（避免自动汇总）
12 router rip
13 version 2
14 no auto-summary
15 ! 宣告直连的192.168.1.0和192.168.3.0网段
16 network 192.168.1.0
17 network 192.168.3.0
18 end
19 write memory
```

R1

```
1 enable
2 configure terminal
3 ! 配置接口Fa0/0的IP地址
4 interface FastEthernet0/0
5 ip address 192.168.2.254 255.255.255.0
6 no shutdown
7 ! 配置接口Fa0/1的IP地址
8 interface FastEthernet0/1
9 ip address 192.168.3.2 255.255.255.0
10 no shutdown
11 ! 启用RIP协议，版本为RIP v2
12 router rip
13 version 2
14 no auto-summary
15 ! 宣告直连的192.168.2.0和192.168.3.0网段
16 network 192.168.2.0
17 network 192.168.3.0
18 end
19 write memory
```

OSPF

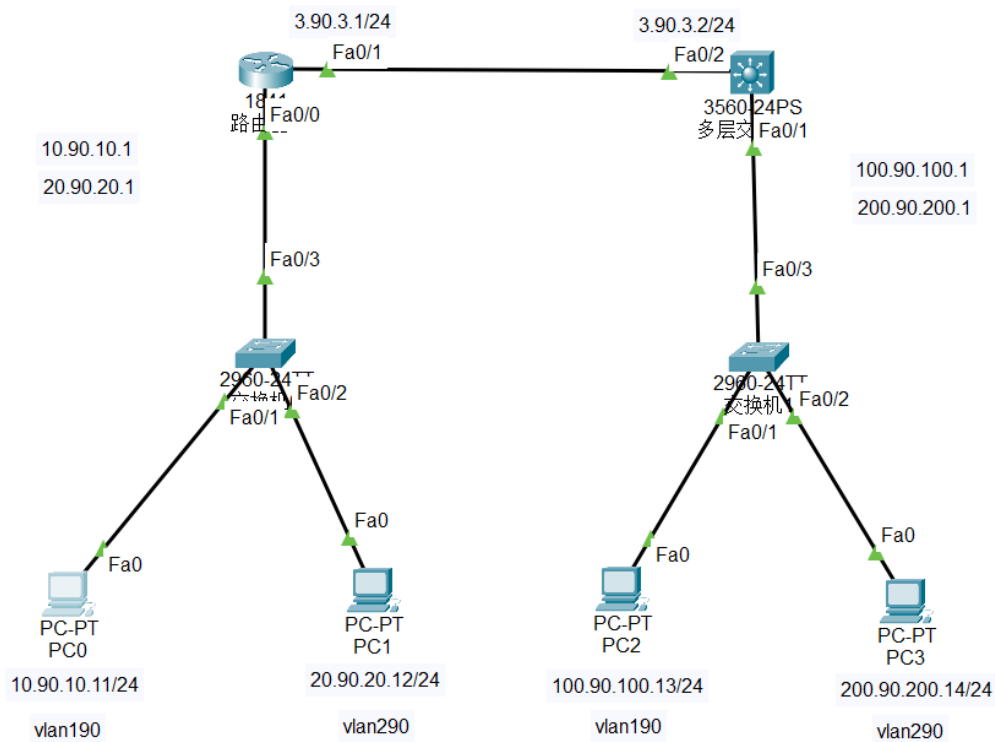
R0

```
1 enable
2 configure terminal
3 ! 配置接口Fa0/0的IP地址
4 interface FastEthernet0/0
5 ip address 192.168.1.254 255.255.255.0
6 no shutdown
7 ! 配置接口Fa0/1的IP地址
8 interface FastEthernet0/1
9 ip address 192.168.3.1 255.255.255.0
10 no shutdown
11 ! 启用OSPF进程（进程号1），将直连网段加入OSPF（区域0）
12 router ospf 1
13 network 192.168.1.0 0.0.0.255 area 0
14 network 192.168.3.0 0.0.0.255 area 0
15 end
16 write memory
```

R1

```
1 enable
2 configure terminal
3 ! 配置接口Fa0/0的IP地址
4 interface FastEthernet0/0
5 ip address 192.168.2.254 255.255.255.0
6 no shutdown
7 ! 配置接口Fa0/1的IP地址
8 interface FastEthernet0/1
9 ip address 192.168.3.2 255.255.255.0
10 no shutdown
11 ! 启用OSPF进程（进程号1），将直连网段加入OSPF（区域0）
12 router ospf 1
13 network 192.168.2.0 0.0.0.255 area 0
14 network 192.168.3.0 0.0.0.255 area 0
15 end
16 write memory
```

实验1-4



左侧二层交换机 接入PC0&1,流通数据给路由器1

```

1  enable
2  configure terminal
3  #创建vlan
4  vlan 190
5  #命名vlan
6  name VLAN190
7  vlan 290
8  name VLAN290
9
10 interface FastEthernet0/1
11 #将模式改为access
12 switchport mode access
13 #接入vlan190
14 switchport access vlan 190
15 #开启端口
16 no shutdown
17
18 interface FastEthernet0/2
19 switchport mode access
20 switchport access vlan 290
21 no shutdown
22
23 #接入路由器 trunk类型
24 interface FastEthernet0/3
25 switchport mode trunk
26 #运行vlan通过
27 switchport trunk allowed vlan 190,290
28 no shutdown

```

单臂路由——路由器 实现PC0&1通信

```

1 enable
2 configure terminal
3 interface FastEthernet0/0
4   no shutdown
5 #配置VLAN190的子接口
6 interface FastEthernet0/0.190
7   encapsulation dot1q 190
8   ip address 10.90.10.1 255.255.255.0 #（网关IP+子网掩码）
9 #配置VLAN290的子接口
10 interface FastEthernet0/0.290
11   encapsulation dot1q 290
12   ip address 20.90.10.1 255.255.255.0
13 #配置互联接口Fa0/1
14 interface FastEthernet0/1
15   ip address 3.90.3.1 255.255.255.0
16   no shutdown

```

右侧二层交换机 接入PC2&3,流通数据给三层交换机

```

1 enable
2 configure terminal
3 #创建vlan
4 vlan 190
5 #命名vlan
6 name VLAN190
7 vlan 290
8 name VLAN290
9
10 interface FastEthernet0/1
11 #将模式改为access
12   switchport mode access
13 #接入vlan190
14   switchport access vlan 190
15 #开启端口
16   no shutdown
17
18 interface FastEthernet0/2
19   switchport mode access
20   switchport access vlan 290
21   no shutdown
22
23 #接入三层交换机（类路由器） trunk类型
24 interface FastEthernet0/3
25   switchport mode trunk
26 #运行vlan通过
27   switchport trunk allowed vlan 190,290
28   no shutdown

```

三层交换机 实现PC2&3之间通信

```

1 enable
2 configure terminal
3
4 ip routing#使三层交换机具有路由器的功能

```

```
5
6 #创建vlan
7 vlan 190
8 #命名vlan
9 name VLAN190
10 vlan 290
11 name VLAN290
12
13
14
15 #配置SVI接口（VLAN的三层网关） 设置vlan190网关
16 interface vlan190
17 ip address 100.90.100.1 255.255.255.0
18 no shutdown
19
20 interface vlan290
21 ip address 200.90.200.1 255.255.255.0
22 no shutdown
23
24 #互联接口Fa0/2
25 interface FastEthernet0/2
26 #开启配置接口IP功能
27 no switchport
28 ip address 3.90.3.2 255.255.255.0
29 no shutdown
```

OSPF——路由器

```
1 router ospf 1
2 network 10.90.10.0 0.0.0.255 area 0
3 network 20.90.20.0 0.0.0.255 area 0
4 network 3.90.3.0 0.0.0.255 area 0
```

OSFP——三层交换机

```
1 router ospf 1
2 network 100.90.100.0 0.0.0.255 area 0
3 network 200.90.200.0 0.0.0.255 area 0
4 network 3.90.3.0 0.0.0.255 area 0
```

最后记得保存 在enable后的界面 写write

```
1 #查看路由表
2 show ip route
3 #查看端口ip配置正确没
4 show ip interface brief
```

